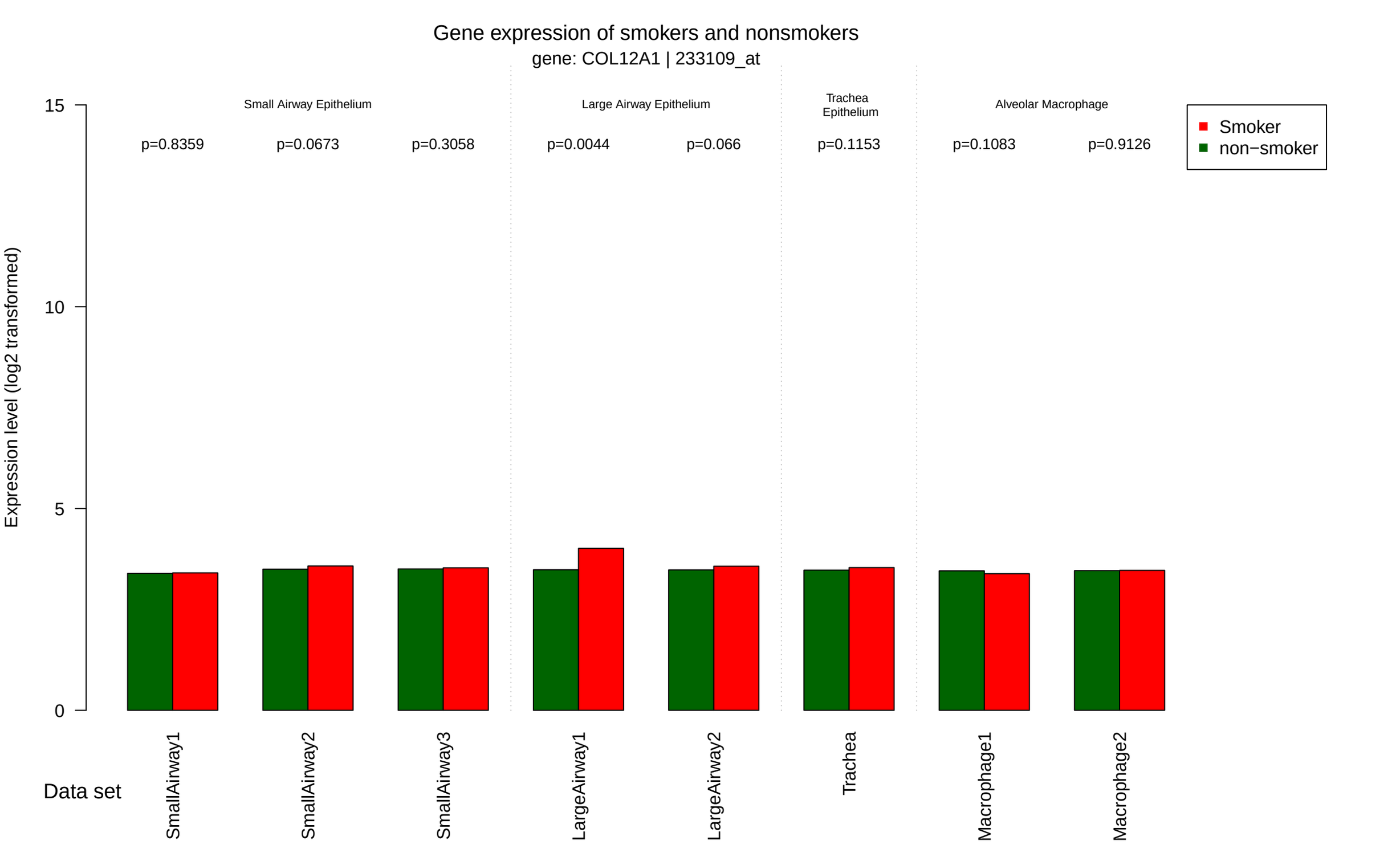
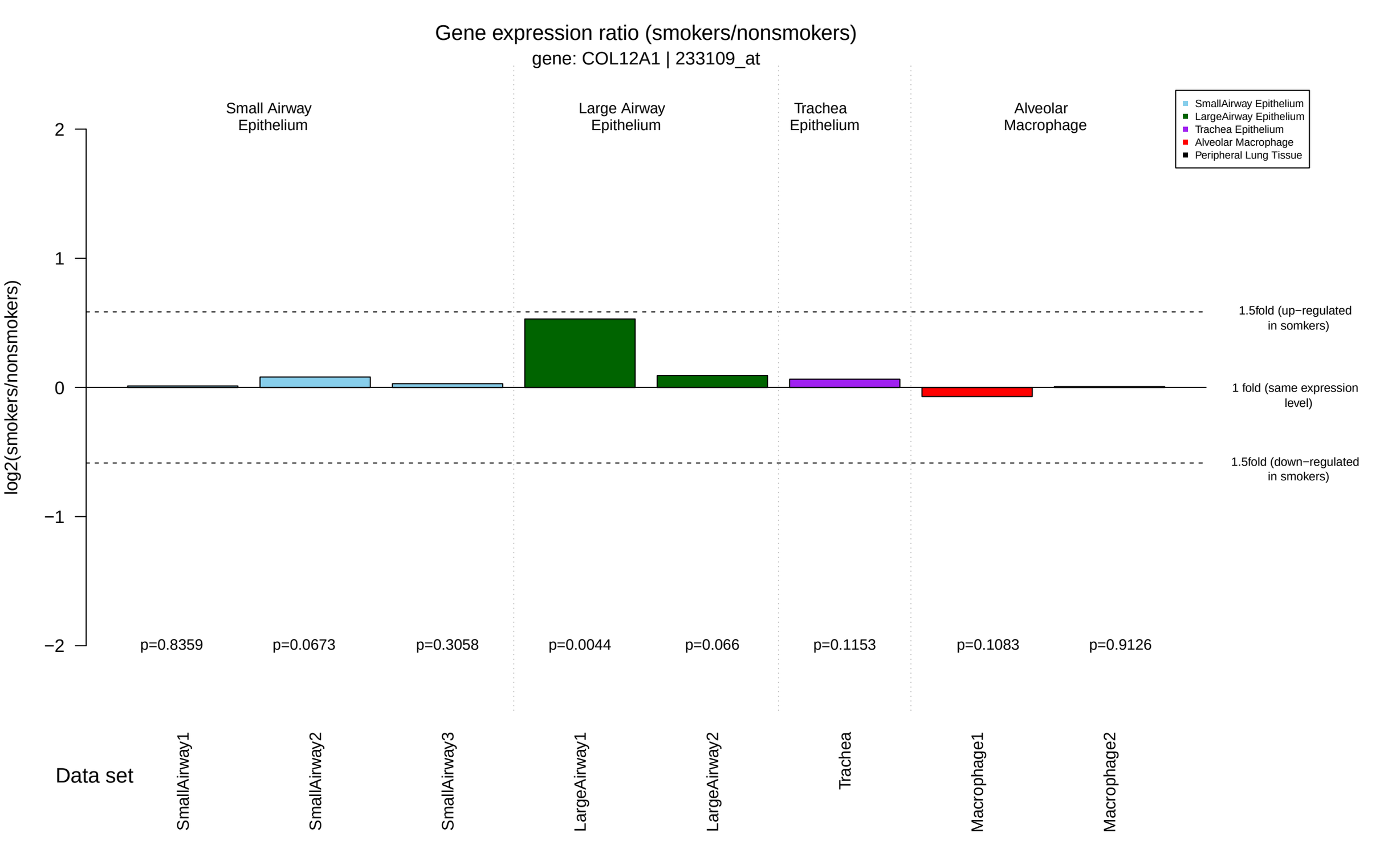
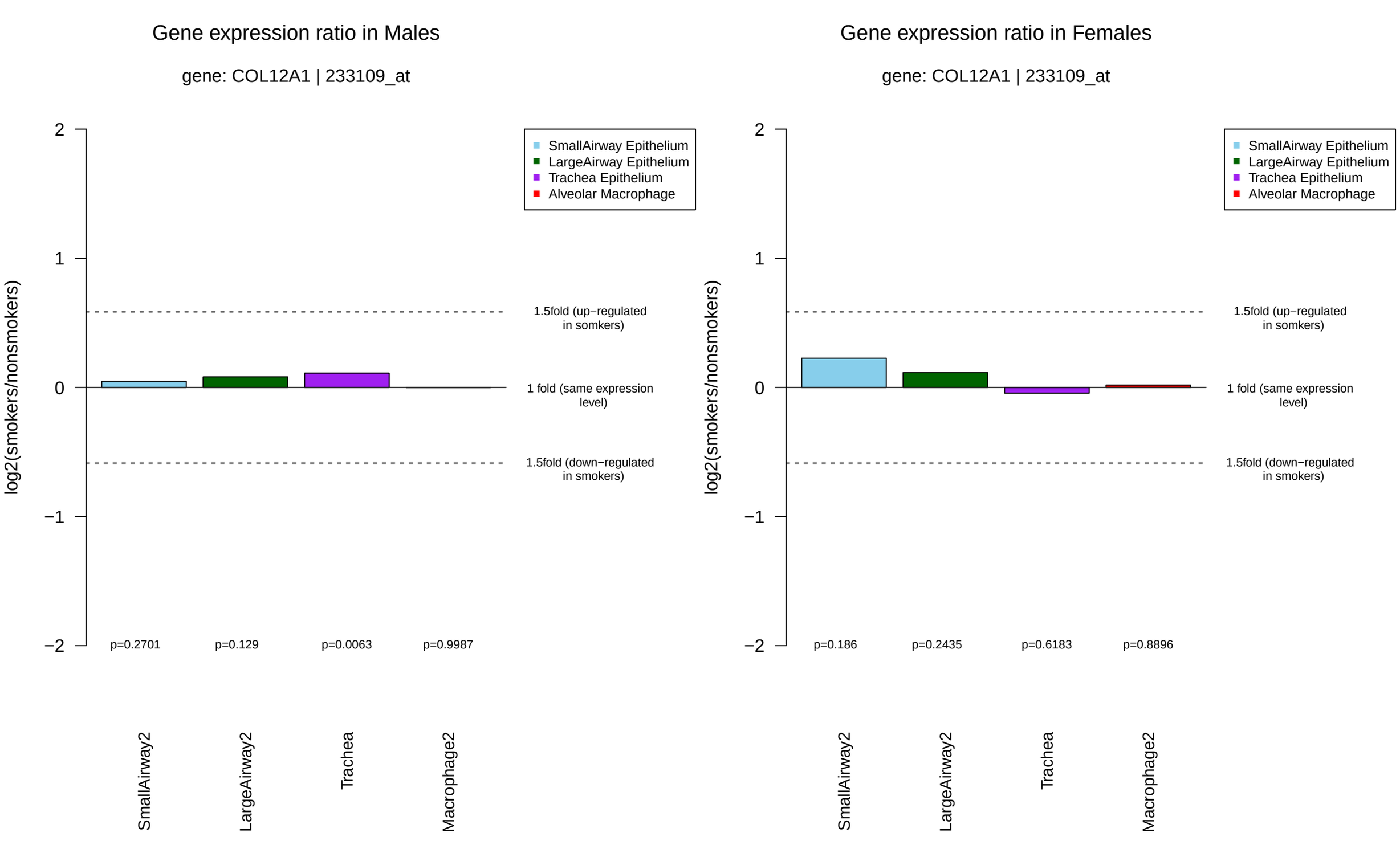
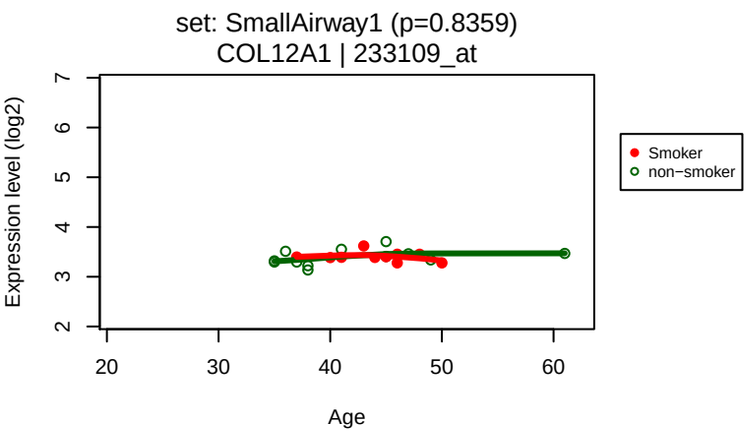
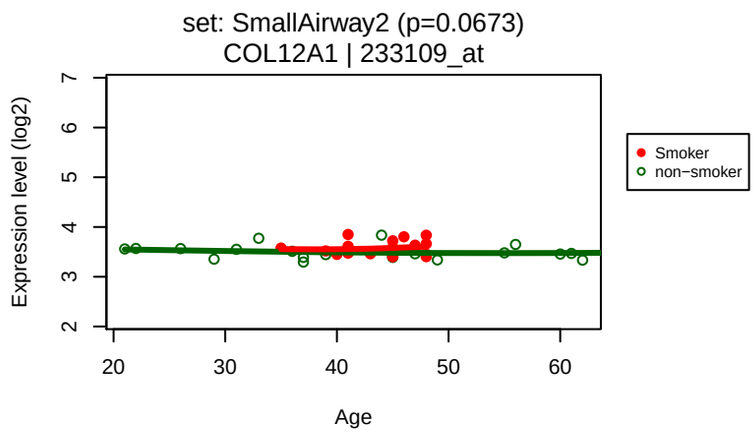
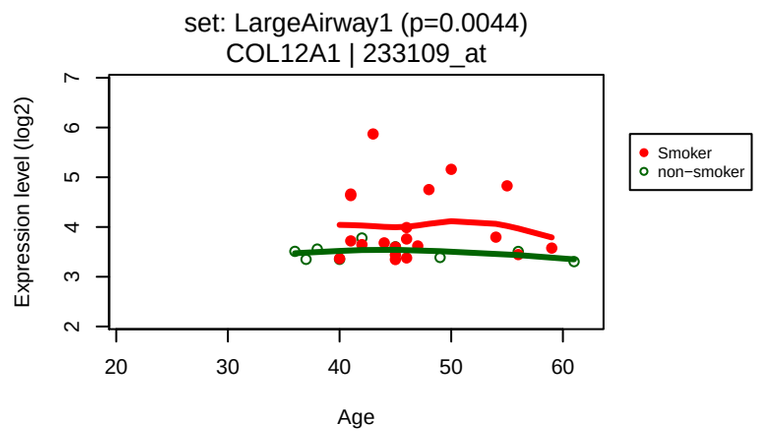

Gene expression of smokers and nonsmokers
gene: COL12A1 | 233109_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8359
p=0.0673
p=0.3058
p=0.0044
p=0.066
p=0.1153
p=0.1083
p=0.9126
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

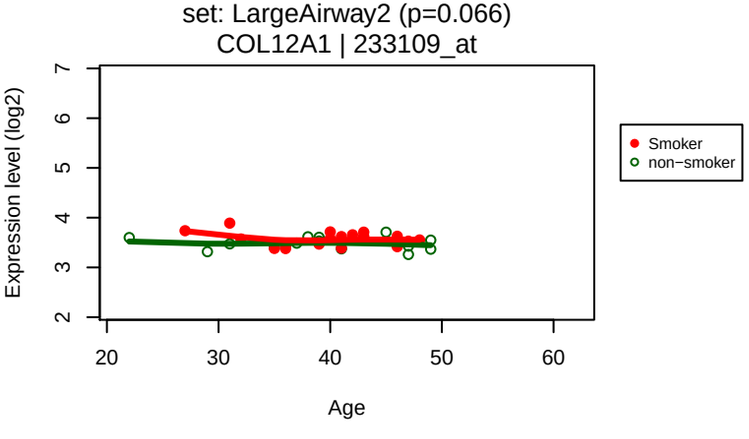

Gene expression ratio (smokers/nonsmokers)
gene: COL12A1 | 233109_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8359
p=0.0673
p=0.3058
p=0.0044
p=0.066
p=0.1153
p=0.1083
p=0.9126
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

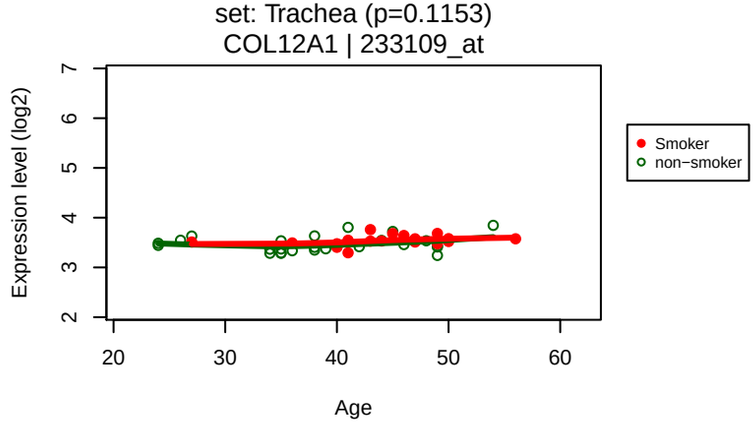

Gene expression ratio in Males
gene: COL12A1 | 233109_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2701
p=0.129
p=0.0063
p=0.9987
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: COL12A1 | 233109_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.186
p=0.2435
p=0.6183
p=0.8896
SmallAirway2
LargeAirway2
Trachea
Macrophage2

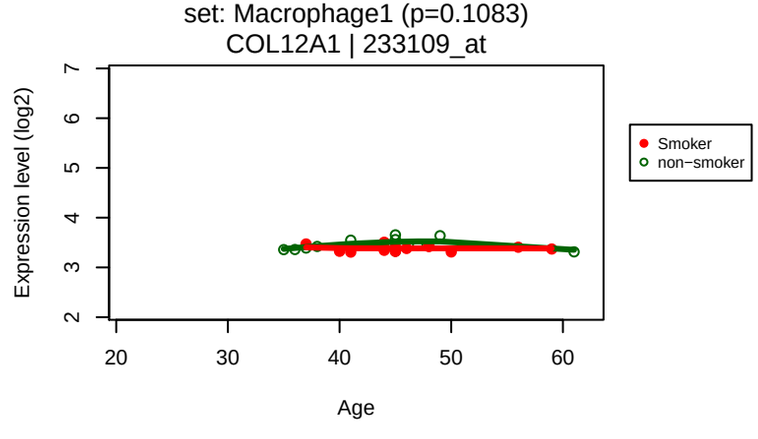

set: SmallAirway1 (p=0.8359)
COL12A1 | 233109_at
Expression level (log2)
Age
Smoker
non−smoker

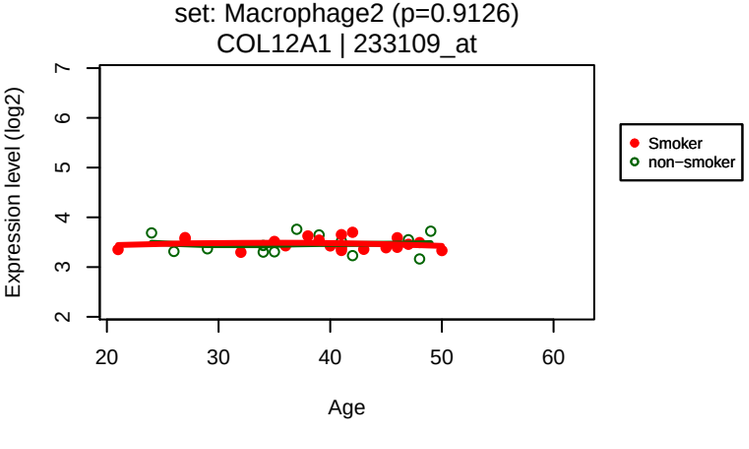

set: SmallAirway2 (p=0.0673)
COL12A1 | 233109_at
Expression level (log2)
Age
Smoker
non−smoker

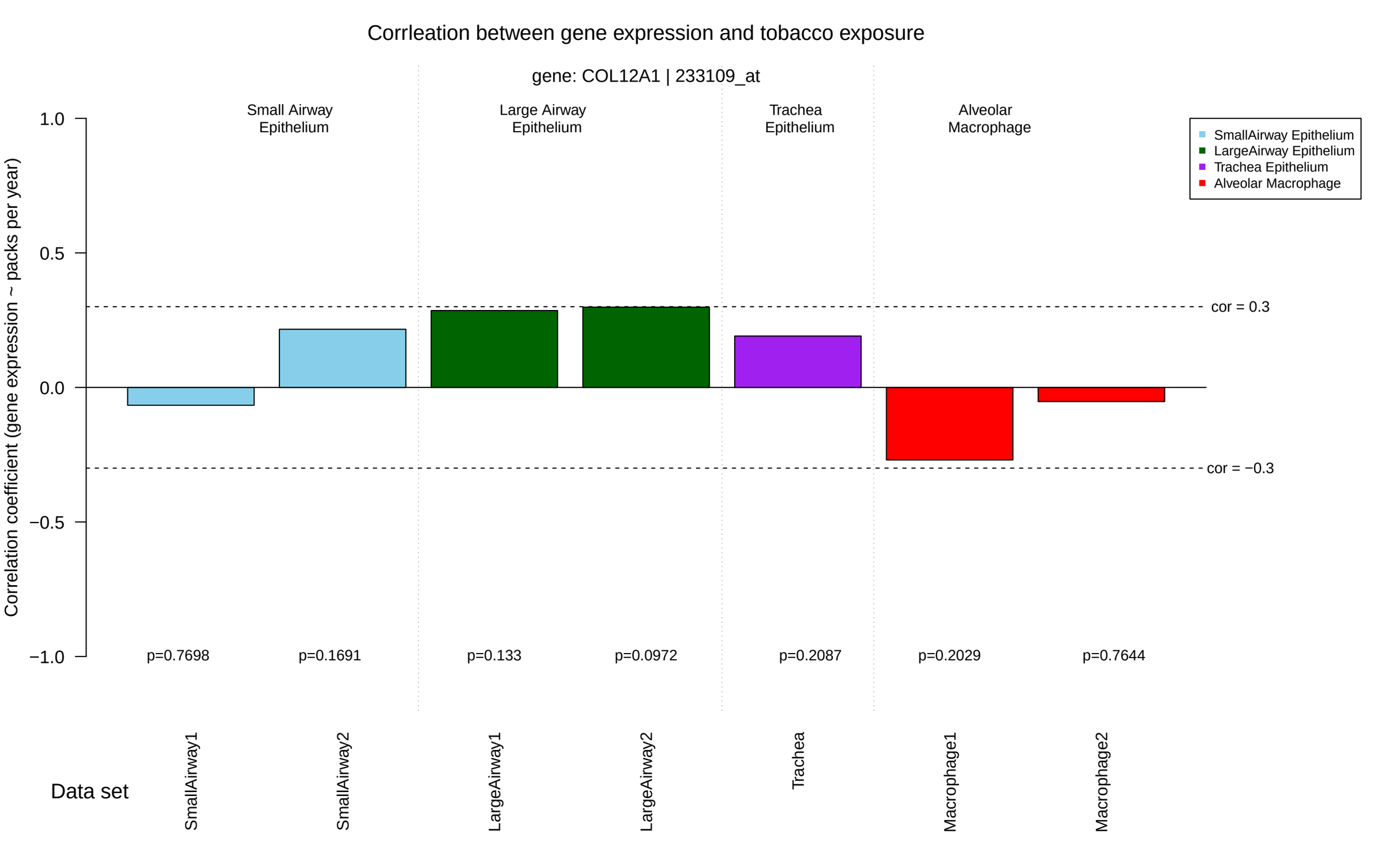

set: LargeAirway1 (p=0.0044)
COL12A1 | 233109_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.066)
COL12A1 | 233109_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1153)
COL12A1 | 233109_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1083)
COL12A1 | 233109_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9126)
COL12A1 | 233109_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: COL12A1 | 233109_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7698
p=0.1691
p=0.133
p=0.0972
p=0.2087
p=0.2029
p=0.7644
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2